

Computing the greatest common divisor of several parametric univariate polynomials via generalized subresultant polynomials

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In this talk, we tackle the following problem: compute the gcd for *several* univariate polynomials with *parametric* coefficients. It amounts to partitioning the parameter space into “cells” so that the gcd has a uniform expression over each cell and constructing a uniform expression of gcd in each cell. We tackle the problem as follows. We begin by making a natural and obvious extension of subresultant polynomials of two polynomials to several polynomials. Then we develop the following structural theories about them.

1. We generalize Sylvester’s theory [1] to several polynomials, in order to obtain an elegant relationship between generalized subresultant polynomials and the gcd of several polynomials, yielding an elegant algorithm.
2. We generalize Habicht’s theory [2] to several polynomials, in order to obtain a systematic relationship between generalized subresultant polynomials and pseudo-remainders, yielding an efficient algorithm.

Using the generalized theories, we present a simple (structurally elegant) algorithm which is significantly more efficient (both in the output size and computing time) than algorithms based on previous approaches.

Keywords

Greatest common divisor, Parametric polynomial, Subresultant polynomial, Sylvester’s theorem, Habicht’s theorem

References

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